Muhammad Talha Atif

Backend Engineer

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SUMMARY

Backend Engineer with 2+ years of production experience building scalable APIs, distributed systems, microservices, and AI-integrated backend platforms. Experienced in Java, Spring Boot, Node.js, Express.js, gRPC, Kafka, PostgreSQL, MongoDB, Redis, and cloud deployment workflows. Delivered compliance-heavy Identity Governance infrastructure across 8+ Spring Boot microservices with event-driven pipelines, secure audit logging, rule-based access automation, and low-latency inter-service communication. Strong focus on backend architecture, database design, performance optimization, system reliability, API security, and clean engineering practices.

WORK EXPERIENCE

Backend Engineer *Dec '25 – Present*
Intimec via MicroTechX Remote

Built backend infrastructure for a compliance-heavy Identity Governance platform, contributing across Spring Boot microservices, audit systems, access automation, gRPC communication, and event-driven backend workflows.

- Migrated internal service-to-service communication from REST to gRPC, reducing inter-service latency by 50% across Auth, Identity, and Access Portal workflows.
- Architected a generic rule-based access engine to assign roles, entitlements, and access items through grouped condition evaluation, CRON scheduling, and event-triggered execution.
- Refactored the audit log schema to remove PII while preserving forensic traceability for compliance, debugging, and security review workflows.
- Fixed a Spring AOP self-invocation issue that was bypassing audit advice on critical backend methods, restoring reliable audit coverage across sensitive operations.
- Contributed to backend architecture across 8+ Spring Boot microservices involving authentication, identity lifecycle management, access requests, logging, audit trails, and inter-service APIs.

Backend Engineer *Sep '25 – Jan '26*
Digital Upgraders LLC Remote

Built and optimized production backend APIs for a social platform serving photographers, videographers, and creative professionals across web and mobile clients.

- Reduced production feed load time from 3s to 700ms by improving query structure, database indexes, response payload size, and caching strategy.
- Fixed duplicate notification delivery by redesigning concurrent write flows with proper database transactions and consistency checks.
- Built and maintained REST APIs for feeds, notifications, user interactions, authentication, media workflows, and creator profile features.
- Led a zero-downtime infrastructure migration from AWS to Fly.io, reworking CI/CD, DNS, service configuration, deployment pipelines, and production environment variables.
- Improved backend reliability by debugging production issues across API responses, database state, deployment configuration, and cross-platform client behavior.

PROJECTS

Moodly | AI-Powered Booking Backend | Demo

- Designed a multimodal emotion detection backend that processes voice recordings and facial images through separate ML classification flows and routes results to a recommendation engine.
- Built RESTful APIs for authentication, booking lifecycle management, mood analysis, host actions, recommendation results, and real-time event updates.
- Implemented an async message queue to decouple media processing from booking recommendations, improving system responsiveness and fault tolerance.
- Added strict input validation, secure authentication, and unit-tested backend business logic across critical API paths.

Memingo | Language Learning Backend Platform | Demo

- Optimized a critical leaderboard API by replacing read-time ranking recalculation with a write-time precomputed cache, reducing response time from 30 seconds to 90ms.
- Built a backend object detection module that receives camera frames, performs item classification, and returns vocabulary plus cultural context as a structured API response.
- Integrated a conversational AI chatbot with session-aware context management, structured output parsing, and personalized learning recommendations based on learner progress.
- Designed backend APIs for lessons, progression tracking, rankings, AI responses, vocabulary data, and user learning history.

SKILLS

Backend Java, Spring Boot, Spring MVC, Spring Security, Spring AOP, Node.js, Express.js, NestJS, FastAPI, Python

APIs REST APIs, gRPC, WebSockets, Swagger, Postman, API Design

Architecture Microservices, Event-Driven Architecture, Distributed Systems, Kafka, RabbitMQ, Message Queues

Databases PostgreSQL, MySQL, MongoDB, SQLite, Redis, Supabase, Firebase, OpenSearch

Security JWT, RBAC, Rate Limiting, Audit Logging

Cloud & DevOps AWS S3, AWS RDS, AWS EC2, Docker, GitHub Actions, CI/CD, Fly.io, Dokploy

AI / ML LangChain, LangGraph, CrewAI, OpenAI SDK, Gemini API, Hugging Face, RAG

AI Workflow Claude, GitHub Copilot, Cursor

Tools Maven, Git, VS Code, IntelliJ IDEA, Stripe

AWARDS

Runner-up, App Development Hackathon (2025) *May '25*
Organized by Punjab University

Ranked, ICPC Asia Topi Online Programming Contest (2024) *Nov '24*
Organized by International Collegiate Programming Contests

EDUCATION

Bachelor's degree in Computer Science, University of Management and Technology – UMT *Oct '21 – Aug '25*